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Illustration 10.12 C++ program to sum the numbers from 1 to 10 using for loop

```
#include <iostream>
using namespace std;
int main ()
{
int i,sum=0;
for(i=1; i<=10;i++)
 {
        sum=sum+i;
 }
cout<<"The sum of 1 to 10 is "<<sum;
return 0;
}
```

Output

The sum of 1 to 10 is 55

Variations of for loop

The for is one of the most important looping statement in C++ because it allows a several variations. These variations increase the flexibility and applicability of for loop. These variations will be discussed below:

Multiple initialization and multiple update expressions

Multiple statements can be used in the initialization and update expressions of for loop. These multiple initialization and multiple update expressions are separated by commas. For example,

```
#include<iostream>
using namespace std;
int main()
{
        int i, j;
            for(i=0, j=10 ;  i<j  ;  i++,j--)
            {
            cout<<"\nThe value of i is"<<i<<" The value of j is "j;
```

Multiple initialization expressions (separated by commas)

Multiple update expressions (separated by commas)